

ITAI-1371 - Intro to Machine Learning
Individual Contribution Journal

Camila Ferreira da Silva

For Lab 04, I combined the group members' work into one final document and reviewed the entire assignment. My contribution focused on organizing the different sections and checking the completed work for consistency and clarity.

Great Inioluwa Olayiwola

I completed a full Exploratory Data Analysis (EDA) on the Titanic dataset, writing and running all code cells for data cleaning, summary statistics, and visualizations using Pandas, Matplotlib, and Seaborn. I also set up two extra experiments to analyze how embarkation ports (Embarked) and ticket prices (Fare) related to survival, uncovering key socioeconomic trends and data skewness. Finally, I wrote detailed answers for the Knowledge Check assessment, explaining the main goals of EDA, defining the highest survival profile (a 1st Class female child), and breaking down why visual plots tell a much deeper story than simple summary stats alone.

Karima Sadruddin

For this lab, my individual work focused on completing and documenting the Exploratory Data Analysis notebook. I reviewed the notebook instructions and followed the workflow from loading the Titanic dataset to examining summary statistics and visualizations. I completed the two student experimentation sections. In the first experiment, I added a count plot to compare survival outcomes by port of embarkation and calculated survival proportions for each port. In the second experiment, I added a box plot to compare ticket fares for passengers who survived and passengers who did not. I included titles and axis labels so the charts would be easier to understand, along with short interpretation statements explaining what to look for. I also completed the three knowledge-check answers in my own words. My answers explain the purpose of EDA, summarize the patterns described by the notebook's class, gender, and age plots, and explain why visualizations are useful alongside summary statistics. I tried to avoid treating a pattern as proof of cause and effect. Another part of my contribution was preparing the reflection and organizing the notebook content for submission. The reflection describes what I learned about using charts and statistics together, selecting a plot for different data types, and questioning assumptions about relationships in data. Remaining verification step: The notebook retrieves the Titanic dataset from an online source. In this working environment, the data URL was not reachable, so I could not verify the chart outputs or claim that every cell executed successfully. I need to open the completed notebook in Google Colab, run all cells, review the results, and export the final executed notebook to PDF. I will make any corrections if the real outputs differ from the expected interpretations. This journal describes the notebook work prepared for this submission. I will edit it if any detail does not match the work I personally completed.

Quan Thach

Experiment 1:

Check for missing values in Embarked (it has a couple of NaNs)

```
print("Missing values in Embarked:",
      df['Embarked'].isnull().sum())

# Count of passengers per port
print(df['Embarked'].value_counts())

# Countplot: survival by port of embarkation
plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df, order=['C',
'Q', 'S'])
plt.title('Survival by Port of Embarkation')
plt.xlabel('Port (C = Cherbourg, Q = Queenstown, S =
Southampton)')
plt.legend(['Died', 'Survived'])
plt.show()

# Raw counts can mislead because the ports had very different
passenger totals,
# so also compute the survival *rate* for each port
survival_by_port =
df.groupby('Embarked')['Survived'].mean().sort_values(ascending=F
alse)
print("\nSurvival rate by port:")
print(survival_by_port.apply(lambda x: f"{x:.1%}"))
```

Experiment 2:

```
# Fare distribution by survival (boxplot)
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.xticks([0, 1], ['Died', 'Survived'])
plt.show()

# Same plot on a log scale, since a few huge fares squash the
boxes
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.yscale('log')
plt.title('Fare Distribution by Survival (log scale)')
```

```
plt.xticks([0, 1], ['Died', 'Survived'])
plt.show()

# Numbers to back up the picture
print(df.groupby('Survived')['Fare'].describe())
print("\nMedian fare:")
print(df.groupby('Survived')['Fare'].median())
```

The main goal of EDA is to understand your data before you model it or draw conclusions from it. That means checking its structure and quality (missing values, odd data types, outliers), finding patterns and relationships between variables, and forming hypotheses worth testing. It's how you figure out which questions to ask and whether the data can answer them. The best survival odds example is a young girl or woman in 1st class. Class, gender, and age all mattered; fare and port mostly echo class. The reverse profile, an adult man in 3rd class, had the worst odds. This matches the "women and children first" pattern, with class layered on top. Fare and port of embarkation are also tracked with survival, but they're likely echoes of class rather than independent effects. Why visualize? Summary stats hide shape, outliers, and group differences. Plots reveal skew, spikes (like young child survivors), and relationships that a mean or count can't. Anscombe's quartet is the classic example. The classic demonstration is **Anscombe's quartet**: four datasets with nearly identical means, variances, and correlations that look completely different when plotted. Summary statistics tell you *what* the center or total is, while plots tell you *what the data actually looks like*.